

Trainings ▾

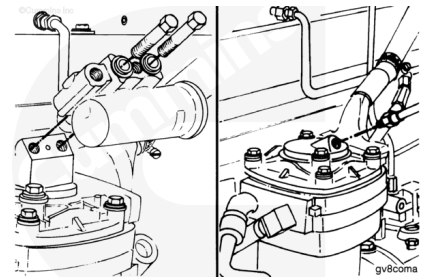
Initial Check

⚠ WARNING ⚠

Air pressure must be released from system before removing the air governor to prevent personal injury.

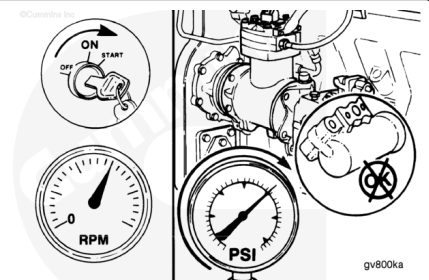
Note : The following procedures are for Holset® air compressors **only**. The illustrations shown will be of the SS model single cylinder air compressor. Differences in procedures for SS, QE and ST model Holset® air compressors will be shown where necessary.

Remove the air governor or air governor hose from the air compressor unloader body.



Operate the engine to activate the air compressor.

If the air compressor is **not** pumping, the unloader valve is malfunctioning and **must** be repaired or replaced.

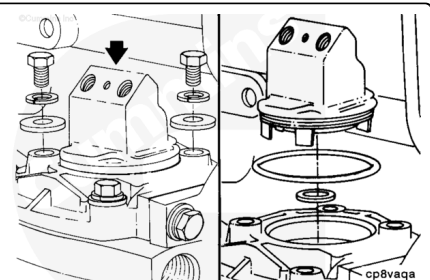


Remove

Holset® SS, E-Type, and ST Models

⚠ WARNING ⚠

The unloader valve body is installed with spring tension. Use care when removing to prevent personal injury. Always wear protective eye wear.



Hold the unloader valve body down and remove the two captive washer capscrews and the two plain washers.

Remove the unloader valve body.

Remove the o-ring seal.

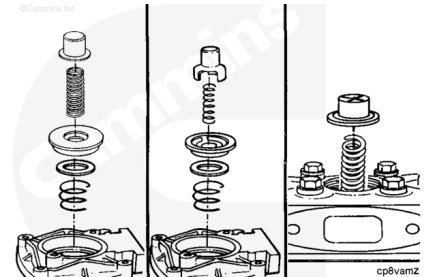
Remove the rectangular ring seal.

Remove the unloader valve cap and the unloader valve spring.

Note : Disassembly of the center unloader valve on Holset® two cylinder air compressors is similar to the single cylinder unloader valve.

Remove the intake valve seat and valve.

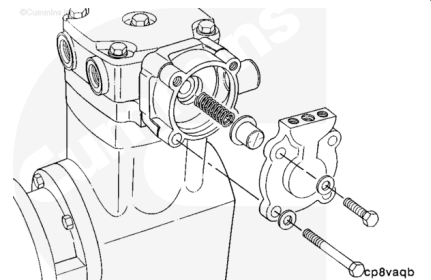
Remove the intake valve spring.



Holset® QE Models

⚠ WARNING ⚠

The unloader body is installed with spring tension. Use care when removing to prevent personal injury. Always wear protective eye wear.

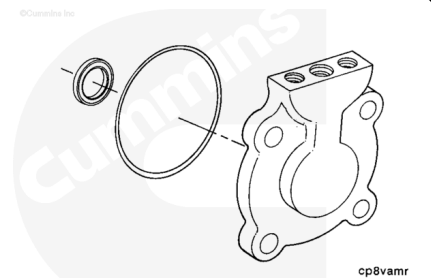


Hold the unloader valve body down and remove the four capscrews.

Remove the unloader valve spring.

Remove the unloader valve cap.

Remove the unloader body gasket and unloader valve cap rectangular ring seal.



Clean and Inspect for Reuse

⚠ WARNING ⚠

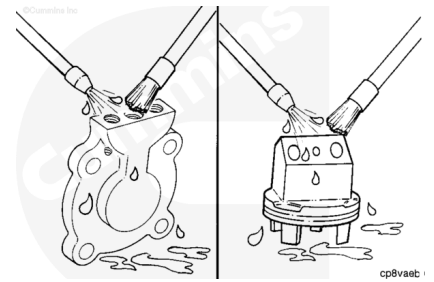
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Do not use caustic cleaners. Caustic cleaners can cause damage to component parts.



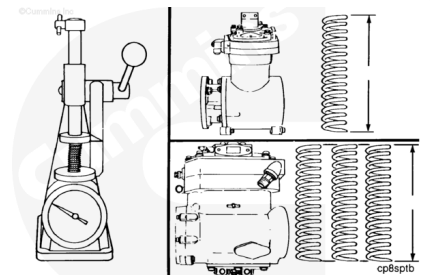
Remove all carbon and varnish from the unloader valve cap body.

Use a valve spring tester to check the unloader spring.

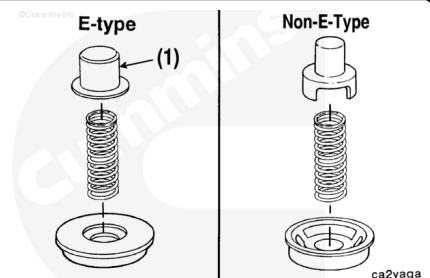
See the Compressor Spring Force Specifications chart shown at the beginning of this section.

Replace the unloader spring if it does **not** meet the specifications shown, or if the wrong spring has been used.

Note : For Holset® two cylinder air compressors, check both cylinder and center unloader springs. Cummins Turbo Technologies™, recommends that new springs be installed during rebuild.



Note : If the compressor has a flat hat type unloader cap (1), it must use an unloader spring and valve seat different than that used with the three prong unloader.



Install

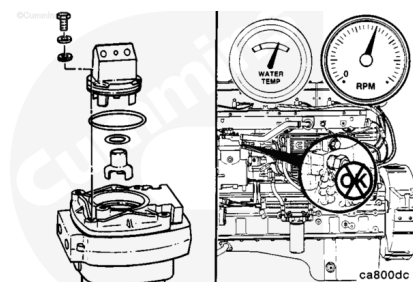
Holset® SS, E-Type, and ST Models

Assemble the air compressor.

Grease rectangular ring seal, unloader cap and unloader body bore with high temperature grease (Accrolube Lubrication Teflon™ grease or equivalent).

Torque Value: 14 n•m [124 in-lb]

Operate the engine and check the compressor for air leaks.

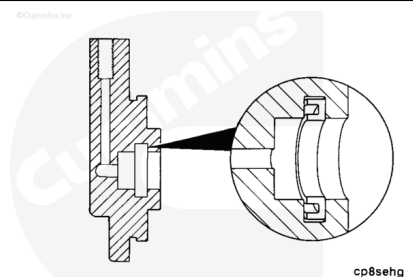


Holset® QE Models

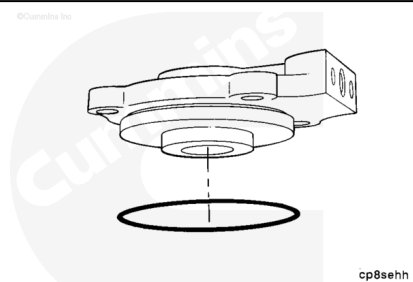
Install the new rectangular V-seal into the unloader body.

Note : The seal must be installed with the grooved side up.

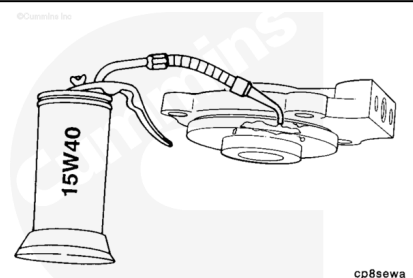
Liberally lubricate the unloader valve bore above and below the rectangular ring seal with high temperature grease (Accrolube Lubrication Teflon™ grease or equivalent).



Install a new o-ring seal on the unloader valve body.



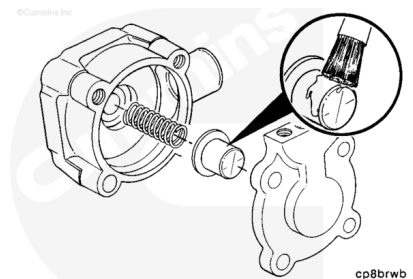
Use clean 15W40 engine oil, Accrolube Lubrication Teflon™ grease, or equivalent to lubricate the seal.



Liberally lubricate the unloader valve body bore and unloader cap with high temperature grease (Accrolube Lubrication Teflon™ grease or equivalent).

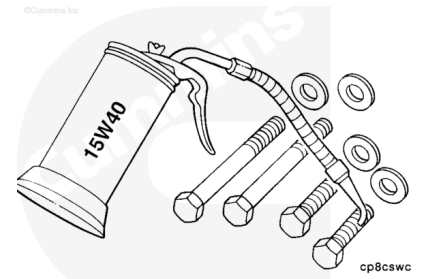
Install the unloader cap.

Install the unloader spring.



Lubricate the unloader screw threads and underhead with clean engine oil (SAE 15W40), before installation.

Note : The two unloader body screws must not be used to attach any brackets.



Assemble the unloader components and attach the unloader assembly to the valve plate with the four capscrews and washers.

Note : The longer capscrews are used to mount the manifold to the air compressor.

Torque Value: 27 n•m [20 ft-lb]

Operate the engine and check the air compressor for air leaks.

